

Tempus Sales Copilot

A Multi-Agent AI Solution for Oncology Territory Intelligence

Case Study · GenAI Product Builder Interview

Northwestern Medicine Territory · 12 Oncologists · 3 Data Sources · 4 Agents

Problem Definition: The Data-to-Insight Gap

The Core Friction

Tempus sales reps spend hours synthesizing disparate data sources before a 15-minute meeting. The ask is to build a copilot that outputs a **ranked provider list**, an **objection handler**, and a **meeting script**.

My Problem Decomposition

Subproblem	My Approach
Data Collection: What data do we need beyond the brief?	Sourced from Northwestern Medicine physician portal, Tempus website, and synthetic CRM notes covering real oncology sales objections
Success Metrics: How do we define a doctor’s impact?	Engineered composite score from likely patient population, Tempus panel fit, and CRM engagement — each representing a distinct dimension of sales opportunity
Technical Implementation: How do we personalize recommendations at scale?	Built with Cursor — a multi-agent pipeline in Streamlit cross-referencing doctor profiles against a scraped Tempus knowledge base
User Interface: How do we surface all three outputs intuitively?	Two-tab UI — ranked territory leaderboard + sales intelligence per doctor

Process Phase 1: Data Collection & Curation

1. Market Intelligence (12 oncologists)

Source: Northwestern Medicine physician portal

- **Specialty Selection:** Queried "oncology" across 396 NM physicians. Used Claude to filter specialties against Tempus's portfolio (xT, xF, xR, xT Heme), retained only the 4 highest-alignment subspecialties.
- **Physician Type:** MDs only; excluded APNs/PAs who lack independent test ordering authority
- **Additional Fields:** Pulled specialty, cancer focus, faculty title, and review count from each NM profile as inputs to the ranking formula
- **Sampling:** Proportional sampling across 4 subspecialties; mixed main campus (Lurie, NMH Chicago) and suburban affiliates (Lake Forest, Delnor, Glenview) to simulate a realistic territory.

2. Product Knowledge Base

Source: Tempus.com – 10 pages scraped

- **Extraction:** Scraped all oncology product pages using Claude (including xT CDx, xF/xF+, xR, xT Heme, algorithmic tests, EHR integration, Tempus Hub, clinical trial matching, digital pathology)
- **Guardrail:** Zero AI-generated content. Every product claim, statistic, and turnaround time is sourced directly from the Tempus website, preventing hallucination in agent outputs

3. CRM Notes

Source: Synthetic

- **Design:** Created 8 physician interaction records covering real oncology sales objections — reimbursement, turnaround time, EHR integration, etc.
- **Intentional Whitespace:** 4 of 12 physicians have no CRM history, replicating authentic territory conditions

Process Phase 2: Feature Engineering & Ranking Logic

Feature Engineering

Since the likely patient population is not readily available in market intelligence, I approximated it using:

$$\text{Patient Population} = 100 + [(60\% \times \text{Review Count}) + (40\% \times \text{Seniority Tier})] * 100$$

60%

Review Count – pulled from Northwestern Medicine. Acts as a proxy for patient-facing activity and prominence within the practice.

40%

Seniority Tier – maps faculty title (Full Professor – High, Associate/Clinical Professor – Med, Assistant Professor – Early) to model institutional influence.

Min-Max Normalization: both proxies are normalized to a 0 to 1 scale to prevent scale skewing.

Range: bounded within 100–200 patients/month, a bracket anchored to the 2012 ASCO Oncologist Survey.

Composite Impact Score

To rank the doctors by potential sales impact, I engineered a weighted composite score using **three proxy signals**:

$$\text{Score} = (40\% \times \text{Likely Patient Population}) + (35\% \times \text{Panel Fit}) + (25\% \times \text{CRM Engagement})$$

40%

Likely Patient Population – feature engineered using Review Count and Seniority Tier

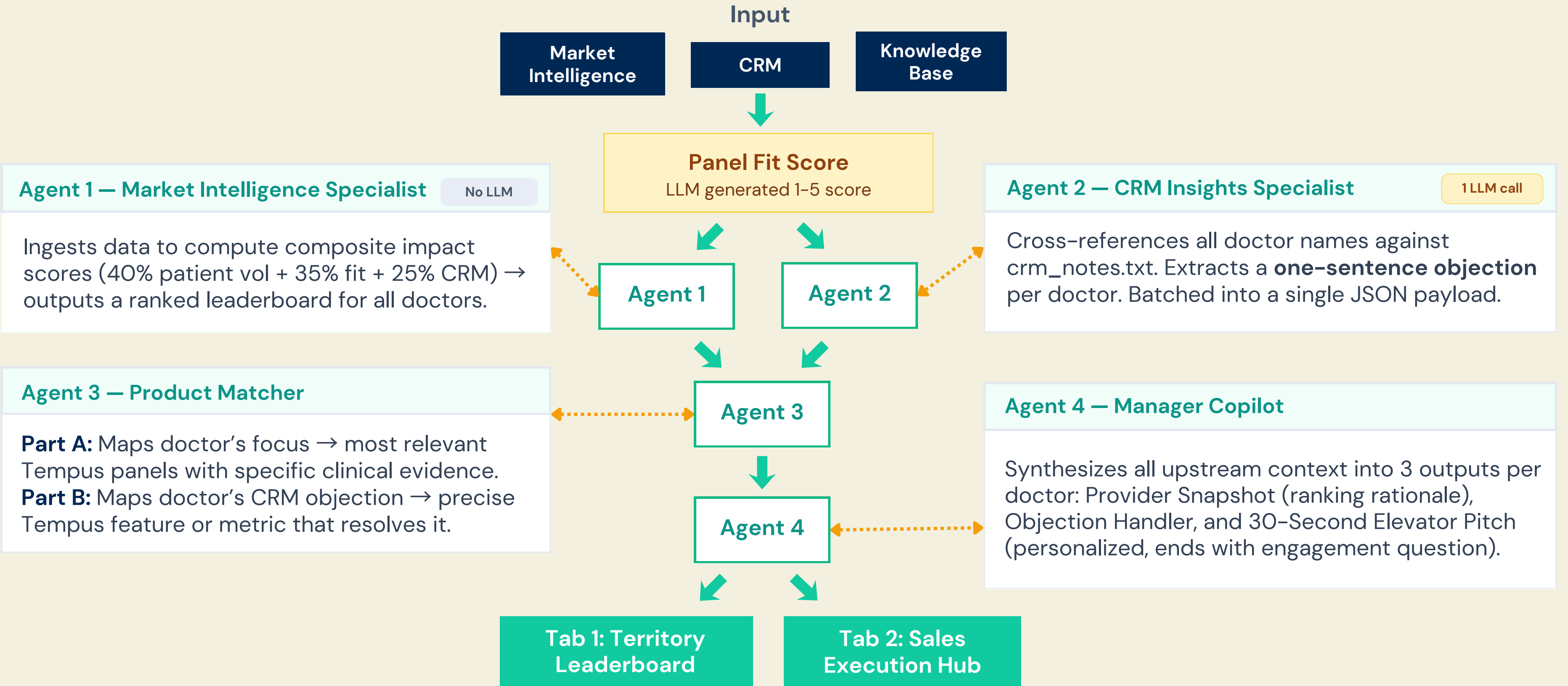
35%

Panel Fit – LLM-scored 1–5 per doctor based on cancer focus alignment to Tempus panels (xT, xF, xR, xT Heme)

25%

CRM Engagement – binary signal: 1 if prior touchpoint of Tempus, 0 if cold territory

Process Phase 3: Multi-Agent Pipeline



Output

Product Solution: Operational Dashboard

A two-tab interface delivering **ranked territory leaderboard** and **personalized sales scripts**.

Tab 1: Ranked Territory Leaderboard

Territory Leaderboard

12 Northwestern Medicine oncologists · Ranked by composite impact score · Click any row to view in Sales Execution Hub

#	PHYSICIAN	SPECIALTY	PRIMARY FOCUS	PTS/MO	FIT	SCORE	CRM
1	Shilajit D. Kundu NMH – Lurie Cancer Center	Urologic Oncology	Prostate Cancer	200	5/5	0.892	Contacted
2	William J. Gradishar NMH – Prentice Women's Hospital	Medical Oncology	Breast Cancer	191	5/5	0.871	Contacted
3	Valerie M. Nelson NM Glenview / Lake Forest Hospi...	Hem/Med Oncology	Breast Cancer	167	4/5	0.782	Contacted
4	Lisa E. Flaum NMH – Prentice Women's Hospital	Hem/Med Oncology	Breast Cancer	166	4/5	0.774	Contacted
5	Raanan Y. Alter NM Winfield / Delnor Hospital	Gynecologic Oncology	Gynecologic Cancers	152	5/5	0.761	Contacted

Doctors ranked by composite score. Top 3 rows highlighted. Click any row to open in Sales Execution Hub.

Tab 2: Sales Execution Hub

Shilajit D. Kundu, MD

Urologic Oncology · NMH – Lurie Cancer Center · High seniority · ✓ CRM on file

AGENTS 1 + 4

Provider Snapshot

#1

Territory rank

200

Patients / mo

0.892

Composite score

Dr. Kundu ranks #1 with the highest composite score of 0.892, driven by the largest patient volume in the territory at 200 patients/month and a perfect panel fit score of 5/5 — his urologic oncology focus maps directly to Tempus's xT CDx and xF panels for prostate, bladder, and kidney cancer. Active CRM history further elevates his priority as a high-impact conversion opportunity.

AGENT 3

Recommended Tempus Products

xT CDx: FDA-approved 648-gene tissue panel identifying HRD status for PARP inhibitor eligibility in prostate cancer, plus TMB and MSI across all GU tumor types. Combined with RNA sequencing, achieves a 43.4% therapy match rate vs 29.6% with DNA alone.

xF Liquid Biopsy: 105-gene ctDNA panel from a blood draw only, results in 7 days. Detects resistance mutations non-invasively and captures unique actionable alterations in 9% of metastatic patients not found in tissue alone — complementing xT CDx rather than replacing it.

Each physician card has: Provider Snapshot · Product Recommendation · Objection Handler · Elevator Pitch

Technical Challenges

The Free-Tier Ceiling Problem

The initial architecture made 25+ individual LLM calls — one per physician per agent. This exhausted every free API tested:

Anthropic CORS restriction blocks browser-side API calls

Groq 12,000 TPM limit + 100,000 daily token cap exhausted during debugging

Mistral Broken v2.4.5 package installation + service capacity exceeded

Google Gemini Free tier quota at 0 for gemini-2.0-flash

Even after JSON batching, daily token limits were exhausted from debugging — requiring the prototype to use pre-computed outputs.

Solution: Two-Track Approach

Architecture Fix: Unified JSON Batching

- Merged Agents 3 + 4 into a single function
- Compressed all 12 physician contexts into one payload, reducing 25+ LLM calls into 3 total
- Eliminated per-provider ThreadPoolExecutor in favor of single batched response
- Reduced API overhead by 80%+

Prototype Reliability Fix: Pre-computation

- All Agent 3+4 outputs were generated beforehand by Claude using the knowledge base, then hardcoded into the prototype
- Ensures zero runtime dependency on external APIs during the demo

Assumptions, Safeguards & Next Steps

Assumptions Made

- **Patient volume proxies:** Review count and seniority correlate with current patient activity, though review count accumulates over career length
- **Composite score weights:** The 40/35/25 weighting is a hypothesis — weights may need adjustment based on Tempus's actual sales motion
- **Panel fit scoring:** LLM-scored 1–5 panel alignment is subjective; different prompts may score differently without ground-truth validation
- **CRM binary signal:** Prior contact treated as equally positive regardless of quality — a negative meeting scores the same as a warm one
- **Synthetic CRM:** Real CRM data would include richer signals (meeting frequency, deal stage, stakeholder map) improving precision

Safeguards Implemented

- **Zero hallucination guardrail:** Agent prompts strictly bounded to scraped KB — all statistics are verified Tempus data
- **Transparent scoring:** Review count and seniority retained as standalone CSV columns — volume estimate is fully auditable

Potential Next Steps

- Connect to **real Salesforce CRM data** via API for live interaction history
- Add a **drug approval monitor** — auto-flag doctors whose cancer focus aligns with newly approved therapies
- **A/B test** composite score weights against actual conversion rates to validate assumptions
- Upgrade to **paid LLM tier** for live pipeline execution without pre-computation